PROVISIONAL PATENT SPECIFICATION

AI AntiHijack

CIPC Submission

Title of the Invention	System and Method for Artificial Intelligence-Based Anti-Hijack Detection, Journey Integrity Monitoring, Secure Vehicle Authorization, Protected Immobilization, Authenticated Reactivation, Multi-Device Correlation, Adaptive Emergency Communication, Predictive Threat Intelligence, Anti-Jamming Communication Resilience, and Tamper-Resilient Evidence Preservation
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Abstract	A system and method for protecting a vehicle and its occupant by learning driver-specific behaviour, monitoring planned, shared, selected, or inferred journeys, correlating signals from an in-vehicle device, a handset, and optionally a wearable device, and detecting anomalous conditions indicative of hijacking, coercion, theft, tampering, signal jamming, or unauthorized startup. The system may determine an authorization state based on one or more trusted credentials and tamper-related signals, place the vehicle in a protected immobilized state when authorization is absent, invalid, or compromised, and permit authenticated reactivation only after verified satisfaction of one or more reactivation conditions. An adaptive communication orchestration engine dynamically selects, ranks, or combines multiple communication pathways including cellular, Bluetooth, Wi-Fi, mesh, sub-GHz, store-and-forward, or delayed-burst channels to maintain emergency capability under degraded or jammed communication conditions. A predictive threat intelligence module analyses historical incident data, hotspot patterns, and behavioural baselines to generate proactive route-risk assessments and pre-journey safety recommendations. A tamper-resilient evidence preservation module cryptographically secures forensic data for post-incident investigation, insurance claim support, and legal proceedings.

Field of the Invention

The present invention relates to vehicle security systems, journey monitoring systems, intelligent authorization systems, and emergency communication platforms. More specifically, the invention

relates to artificial intelligence systems capable of learning driver behaviour, monitoring journey integrity, correlating signals from a vehicle, a mobile handset, and optionally a wearable device, detecting unauthorized startup, signal jamming, or tampering conditions, and initiating adaptive anti-hijack and emergency response actions.

Definitions

- **Vehicle Device:** An in-vehicle electronic device installed in or associated with a vehicle and configured to collect, process, transmit, or store vehicle-related data.
- **Driver Behaviour Profile:** A data-driven representation of normal or expected behaviour of a specific driver, including route choices, timing patterns, speed tendencies, stop tendencies, acceleration tendencies, or any combination thereof.
- **Threat Score:** A computed risk value derived from one or more contextual signals and used to classify the likelihood or severity of distress, hijack, tamper, theft, coercion, unauthorized activity, or signal interference.
- **Communication Orchestration Engine:** A computational module configured to evaluate available communication pathways and select, rank, combine, or sequence pathways for one or more transmissions.
- **Authorization State:** A machine-determined state indicating whether one or more valid authorization conditions exist for vehicle startup, operability, reactivation, or continued use.
- **Trusted Credential:** A key transponder, OEM immobilizer credential, secure token, paired handset, paired wearable, cryptographic secret, server-issued authorization code, biometric confirmation, or other credential recognised as satisfying at least part of an authorization condition.
- **Protected Immobilized State:** A recoverable secured state in which startup or selected vehicle functions are inhibited, while preserving tamper records, alert functionality, or reactivation controls.
- **Authenticated Reactivation:** A controlled process by which a protected immobilized state is removed after verified presentation of one or more trusted credentials, tokens, confirmations, or service authorizations.
- **Predictive Threat Intelligence:** Analytical insights generated from historical incident data, geographic crime patterns, temporal risk profiles, and driver behavioural baselines, used to forecast elevated risk conditions before they materialise.
- **Tamper-Resilient Evidence Record:** A cryptographically secured, timestamped record of forensic data including location breadcrumbs, audio captures, tamper logs, and communication attempts, preserved for post-incident investigation, insurance claims, or legal proceedings.
- **Signal Interference Event:** A condition in which one or more communication or location channels are degraded, denied, jammed, spoofed, or rendered unreliable beyond normal environmental variation.
- **Anti-Jamming Communication Resilience Layer:** A subsystem configured to detect or infer signal interference, preserve local decision capability, switch among primary and fallback communication paths, perform store-and-forward or delayed-burst transmission, and preserve forensic records during degraded connectivity.

Background of the Invention

Vehicle hijacking, coercive vehicle diversion, vehicle theft, signal interference, and unauthorized startup remain significant safety and asset-protection risks in many regions. Conventional vehicle tracking systems typically rely on GPS and cellular communication, often require manual panic-button activation, and frequently become less effective when a driver cannot safely interact with a device or when a communication jammer is present.

Known immobilizer and key-based systems generally verify only a limited startup credential and may not correlate route context, vehicle telemetry, trusted-device presence, or startup-sequence anomalies to determine whether a broader threat condition exists.

Existing systems also do not provide predictive threat intelligence based on historical crime data correlated with driver-specific behavioural patterns, nor do they generate tamper-resilient forensic evidence packages suitable for insurance claim substantiation and legal proceedings.

Summary of the Invention

- an in-vehicle electronic device, an artificial intelligence processor, a journey-monitoring module, and a communication module configured for one or more communication pathways;
- driver-specific behavioural learning and journey integrity monitoring for planned, shared, selected, or inferred journeys;
- multi-device correlation across vehicle, handset, wearable, key, route context, and optional cloud inputs;
- a secure authorization layer configured to determine an authorization state and to place the vehicle in a protected immobilized state when authorization is absent, invalid, or compromised;
- authenticated reactivation only after verified satisfaction of one or more reactivation conditions;
- predictive threat intelligence using historical incident data, temporal risk windows, and route-risk forecasting;
- an anti-jamming communication resilience layer configured to detect signal interference, preserve local decision support, and switch among primary and fallback pathways including store-and-forward or delayed-burst transmission;
- a tamper-resilient evidence preservation module configured to cryptographically secure forensic data and generate incident packages.

System Overview

Module	Purpose
In-vehicle AI device	Local capture, processing, tamper sensing, and decision support
Driver and trusted-contact applications	Journey setup, trip sharing, event review, escalation visibility
Communication orchestration engine	Adaptive multi-path emergency communication and failover
Secure authorization layer	Startup validation, protected lockout, authenticated reactivation
Predictive threat intelligence module	Pre-journey risk scoring, route recommendations, dynamic sensitivity
Anti-jamming communication resilience layer	Interference detection, path failover, delayed transmission, evidence preservation
Evidence preservation module	Cryptographic evidence capture, sync, reconstruction, claim support

Brief Description of the Drawings

- Figure 1 illustrates an example overall platform architecture.
- Figure 2 illustrates an example communication architecture with multiple fallback channels.
- Figure 3 illustrates an example artificial intelligence detection pipeline.
- Figure 11 illustrates an example startup authorization, protected lockout, and authenticated reactivation flow.
- Figure 12 illustrates an example predictive threat intelligence pipeline.
- Figure 13 illustrates an example tamper-resilient evidence preservation and forensic reconstruction workflow.
- Figure 14 illustrates an example anti-jamming communication resilience workflow.

Detailed Description of the Invention

1. In-Vehicle AI Device

The system includes a concealed or otherwise installed in-vehicle electronic device configured to capture location, motion, telemetry, and communication-related data. The device may be OEM integrated, ECU associated, or implemented as a hardened secondary module. The device may include secure storage, one or more radios, one or more sensors, optional backup power, and one or more interfaces to vehicle systems including CAN, OBD, or equivalent interfaces.

2. Driver Behaviour Learning

The artificial intelligence processor records historical journey data and constructs driver-specific behavioural profiles. Learning may include route preferences, departure timing, arrival timing, stop tendencies, braking style, acceleration style, typical trip durations, and normal contexts of use. Profiles may be maintained separately for multiple authorised drivers.

3. Journey Monitoring and Live Route Following

A driver may define an intended journey in a mobile application, including home, school, work, or a custom route, or the system may infer an expected route from prior behaviour. Real-time movement may be compared with the expected route or destination progress in order to classify a journey state.

4. Communication Orchestration

The communication orchestration engine evaluates available communication pathways and selects, sequences, or combines one or more pathways according to expected reliability, latency, stealth, power cost, and threat context. Pathways may include cellular communication, Bluetooth relay to nearby devices, Wi-Fi offload, mesh or sub-GHz transmission, store-and-forward delivery, or cloud-assisted routing.

5. Secure Authorization and Protected Immobilization Layer

The platform may determine an authorization state for startup, operability, or reactivation based on one or more trusted credentials and one or more anomaly or tamper indicators. When the authorization state is absent, invalid, or compromised, the platform may place the vehicle in a protected immobilized state in which startup or selected vehicle functions are inhibited while tamper logs, alert capability, and reactivation controls are preserved. The protected immobilized state is preferably recoverable rather than destructive.

6. Authenticated Reactivation Workflow

The platform may permit authenticated reactivation only after verified satisfaction of one or more reactivation conditions. Such conditions may include return of a valid credential, owner confirmation, trusted-device confirmation, service-agent confirmation, server-side approval, biometric confirmation, one-time token validation, or a combination thereof. Reactivation events are preferably logged and previously used reactivation tokens are invalidated after use.

7. Predictive Threat Intelligence Module

The system may aggregate anonymised historical vehicle crime data by geographic area, time of day, day of week, and incident type, and correlate that data with a driver's planned or inferred route. The module may generate pre-journey risk assessments, identify temporal risk windows, recommend lower-risk alternative routes, and increase detection sensitivity before an incident occurs.

8. Anti-Jamming Communication Resilience Layer

The system may monitor communication quality, signal strength, signal-to-noise ratio, GNSS availability, radio health, or channel consistency to detect or infer a signal interference event. When suspected jamming, denial, or spoofing is detected, the platform may preserve local decision capability, continue anomaly scoring, preserve forensic data, perform dead-reckoning from motion or vehicle-bus data, and switch or sequence transmissions across available pathways including cellular, Bluetooth relay, Wi-Fi, mesh, sub-GHz, store-and-forward, or delayed-burst transmission. The system may transmit immediately over an available primary or fallback channel, or may defer transmission until a temporary connectivity window is detected.

9. Tamper-Resilient Evidence Preservation

The system may autonomously capture, hash, timestamp, and preserve forensic data including location breadcrumbs, audio, tamper logs, communication attempts, and sensor readings. Evidence may be stored locally, synchronised to secure cloud storage using end-to-end encryption, or both. The system may generate a structured incident package suitable for insurance claims and legal or investigative use.

10. Best Method of Performing the Invention

In a preferred embodiment, the invention is implemented as an OEM-integrated or hardened in-vehicle security module with secure storage, local AI inference, backup power, CAN or OBD integration, and multiple radios. The preferred authorization workflow uses at least one trusted credential together with startup-sequence analysis and tamper indicators to determine an authorization state, after which the vehicle either enters a normal operable state or a protected immobilized state. The preferred reactivation workflow uses one or more verified credentials and a logged, recoverable token-based reactivation process. The preferred evidence workflow uses cryptographic hashing and timestamping with opportunistic secure cloud synchronisation. The preferred anti-jamming workflow uses local interference detection, fallback-path selection, dead-reckoning during GNSS loss, and opportunistic store-and-forward or delayed-burst alert transmission.

11. Industrial Applicability

The invention is industrially applicable to private passenger vehicles, high-risk commuter vehicles, family safety deployments, school transport, fleet management, insurer-linked telematics, rental vehicles, logistics fleets, and security-company supported response platforms. The invention may be implemented wholly on-device, wholly in a remote platform, or in hybrid form.

Example Embodiments

- A startup attempt occurs in the absence of a valid trusted credential and in the presence of an abnormal startup sequence on the vehicle bus. The authorization state is determined to be invalid, the platform places the vehicle in a protected immobilized state, and one or more tamper alerts are sent. Subsequent authenticated reactivation is permitted only after verification of a trusted credential and a valid reactivation condition.
- The system analyses historical crime data and identifies that a driver's regular evening commute route passes through an intersection with elevated hijacking risk between 18:00 and 20:00. The system generates a pre-journey notification recommending an alternative route during those hours and increases detection sensitivity when the driver approaches that area.
- During a hijacking incident, the system captures vehicle location breadcrumbs at increased frequency, records ambient audio, logs tamper events and communication attempts, cryptographically secures all captured data, and synchronises the evidence package to secure cloud storage.
- A signal jammer causes simultaneous GNSS degradation and cellular collapse while vehicle motion sensors indicate continued movement. The system flags a signal interference event, switches to a fallback pathway, stores forensic records locally, derives interim movement estimates from motion or bus data, and transmits delayed alerts once a usable communication window is detected.

Advantages of the Invention

- improved early detection of hijack or coercive diversion by combining behavioural learning with route monitoring
- continued emergency capability under degraded communication conditions through adaptive pathway selection
- support for protected immobilization and authenticated reactivation in unauthorized startup or tamper scenarios
- proactive threat avoidance through predictive intelligence and dynamic route recommendations
- tamper-resilient forensic evidence with cryptographic integrity for insurance and legal proceedings

- enhanced resilience against communication jamming through interference detection, fallback channels, and delayed-burst recovery

Claims

1. 1. A vehicle security platform comprising: an in-vehicle electronic device; an artificial intelligence processor associated with the in-vehicle electronic device; a journey-monitoring module; and a communication module configured for one or more communication pathways, wherein the artificial intelligence processor is configured to detect anomalous or distress-related conditions associated with a vehicle journey.
2. 2. The platform of claim 1, wherein the artificial intelligence processor is configured to learn driver-specific behavioural patterns from historical driving data.
3. 3. The platform of claim 1 or claim 2, wherein the platform is configured to compare real-time vehicle movement with an intended, selected, shared, or inferred journey and to classify route deviations according to context.
4. 4. The platform of any one of the preceding claims, wherein the platform computes a threat score using two or more contextual signals.
5. 5. The platform of any one of the preceding claims, further comprising a communication orchestration engine configured to select, sequence, combine, or rank multiple communication pathways.
6. 6. The platform of any one of the preceding claims, further comprising a wearable-device integration layer configured to receive one or more wearable-device signals and correlate the wearable-device signals with vehicle route data and handset status data.
7. 7. The platform of any one of the preceding claims, wherein a voice-recognition module detects a predefined distress phrase and initiates covert emergency mode.
8. 8. The platform of any one of the preceding claims, wherein the response policy engine executes a staged escalation sequence including increased monitoring, trusted-contact alerts, covert tracking, or responder escalation.
9. 9. A vehicle startup authorization and immobilization system comprising: a secure authorization layer configured to determine an authorization state for vehicle startup, operability, lockout, or reactivation based on one or more trusted credentials and one or more anomaly or tamper indicators; and a control module configured, when the authorization state is absent, invalid, or compromised, to place the vehicle in a protected immobilized state in which startup or one or more selected vehicle functions are inhibited while alerting, logging, or monitoring capabilities are maintained.
10. 10. The system of claim 9, wherein the trusted credentials include one or more of a key transponder, an OEM immobilizer credential, a handset token, a wearable token, a cryptographic secret, a biometric confirmation, a server-issued code, or a one-time authorization token.
11. 11. The system of claim 9 or claim 10, wherein authenticated reactivation requires verification of one or more reactivation conditions including presentation of a valid trusted credential, owner confirmation, service-agent confirmation, trusted-device confirmation, server-side authorization, or one-time token validation.
12. 12. A predictive vehicle threat intelligence system comprising: a predictive threat intelligence module configured to generate proactive risk assessments by correlating historical incident data, geographic crime patterns, temporal risk profiles, and driver behavioural baselines.
13. 13. The system of claim 12, wherein the predictive threat intelligence module generates pre-journey risk notifications and alternative route recommendations when elevated risk is identified on a planned or inferred route.
14. 14. The system of claim 12 or claim 13, wherein the predictive threat intelligence module identifies recurring temporal risk windows and autonomously adjusts detection sensitivity during those windows.

15. 15. A tamper-resilient forensic evidence preservation system comprising: an evidence capture module configured to capture forensic data from multiple sources during a vehicle security incident; an integrity module configured to cryptographically hash and timestamp the forensic data to establish chain-of-custody integrity; and a reconstruction module configured to generate a structured incident package suitable for insurance claims, investigation, or legal proceedings.
16. 16. The system of claim 15, wherein the forensic data includes one or more of vehicle telemetry, handset location history, wearable data, communication logs, audio captures, tamper event logs, and sensor readings.
17. 17. The system of claim 15 or claim 16, wherein evidence packages are synchronised to secure cloud storage using end-to-end encryption and opportunistic connectivity windows.
18. 18. An anti-jamming communication resilience system comprising: an interference detection module configured to detect or infer a signal interference event from degradation in one or more communication or location channels; and a failover module configured to preserve local decision support and to select, sequence, combine, or defer one or more transmissions over available primary or fallback pathways including cellular, Bluetooth, Wi-Fi, mesh, sub-GHz, store-and-forward, or delayed-burst channels.
19. 19. The system of claim 18, wherein the interference detection module correlates radio degradation with motion, route, or telemetry data to distinguish signal interference from ordinary signal loss.
20. 20. The system of claim 18 or claim 19, wherein the system preserves a local evidence record and performs dead-reckoning from motion or vehicle-bus inputs during at least part of a signal interference event.
21. 21. A computer-implemented method of monitoring a vehicle journey, the method comprising: learning one or more behavioural patterns associated with a driver; comparing live vehicle movement with an intended, selected, shared, or inferred journey; computing a threat score from two or more contextual signals; and executing one or more response actions according to the threat score.
22. 22. A computer-implemented method of controlling vehicle startup authorization, the method comprising: determining an authorization state from one or more trusted credentials and one or more anomaly or tamper indicators; inhibiting startup or placing the vehicle into a protected immobilized state when the authorization state is absent, invalid, or compromised; and permitting authenticated reactivation only after verified satisfaction of one or more reactivation conditions.
23. 23. A computer-implemented method of generating predictive vehicle threat intelligence, the method comprising: aggregating historical vehicle crime data by geographic area, time, and incident type; correlating aggregated data with a driver's planned or inferred route; generating a pre-journey risk assessment; and adjusting detection sensitivity based on the predicted risk profile.
24. 24. A computer-implemented method of preserving tamper-resilient forensic evidence during a vehicle security incident, the method comprising: capturing forensic data from multiple vehicle and device sensors; cryptographically hashing and timestamping captured data to establish integrity; correlating multi-source data to construct an incident timeline; and generating a structured evidence package suitable for insurance claims or legal proceedings.
25. 25. A computer-implemented method of maintaining emergency communication capability during a signal interference event, the method comprising: detecting or inferring degradation of one or more communication or location channels; preserving local anomaly evaluation and evidence capture during the degradation; selecting or sequencing one or more fallback communication pathways; and transmitting, storing, or deferring one or more alerts or evidence packages until a transmission condition is satisfied.
26. 26. A non-transitory computer-readable medium carrying instructions which, when executed by one or more processors of an in-vehicle device, handset, wearable device, or remote platform, cause performance of the method of any one of claims 21 to 25.
27. 27. A vehicle security system substantially as herein described with reference to the accompanying brief and drawings.

Drawings

Figure 1 - Overall platform architecture.

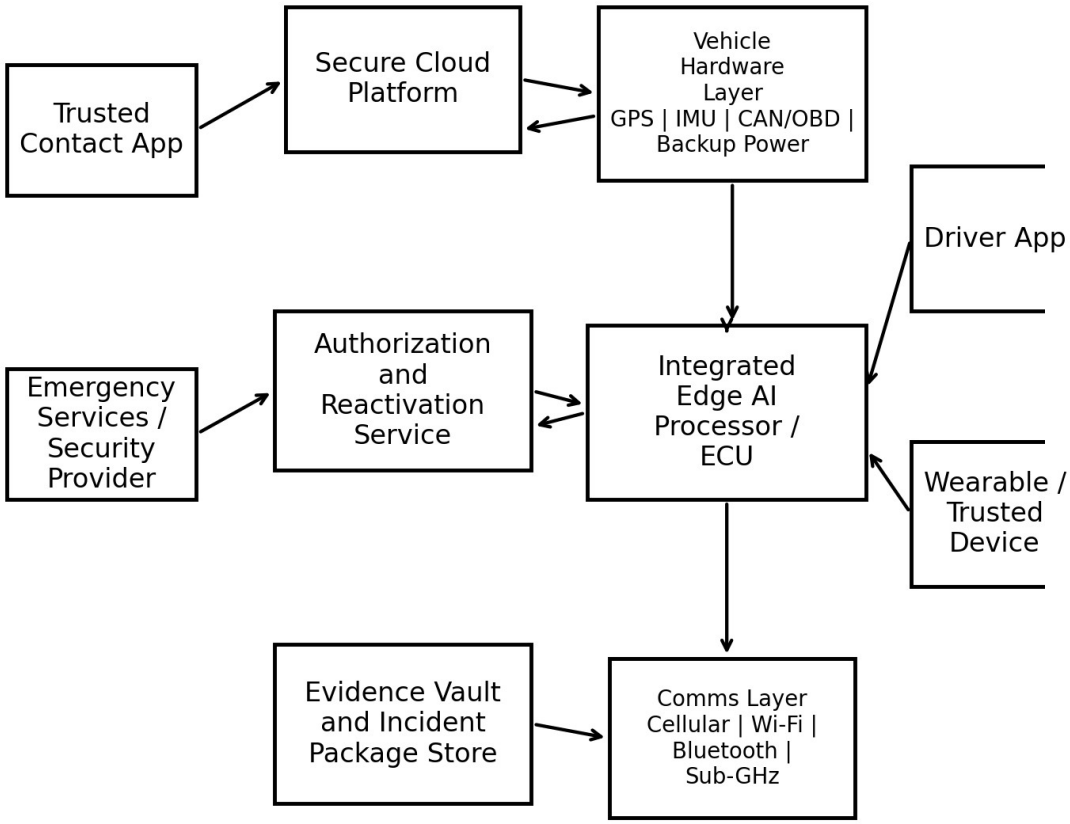


Figure 2 - Communication architecture with multiple fallback channels.

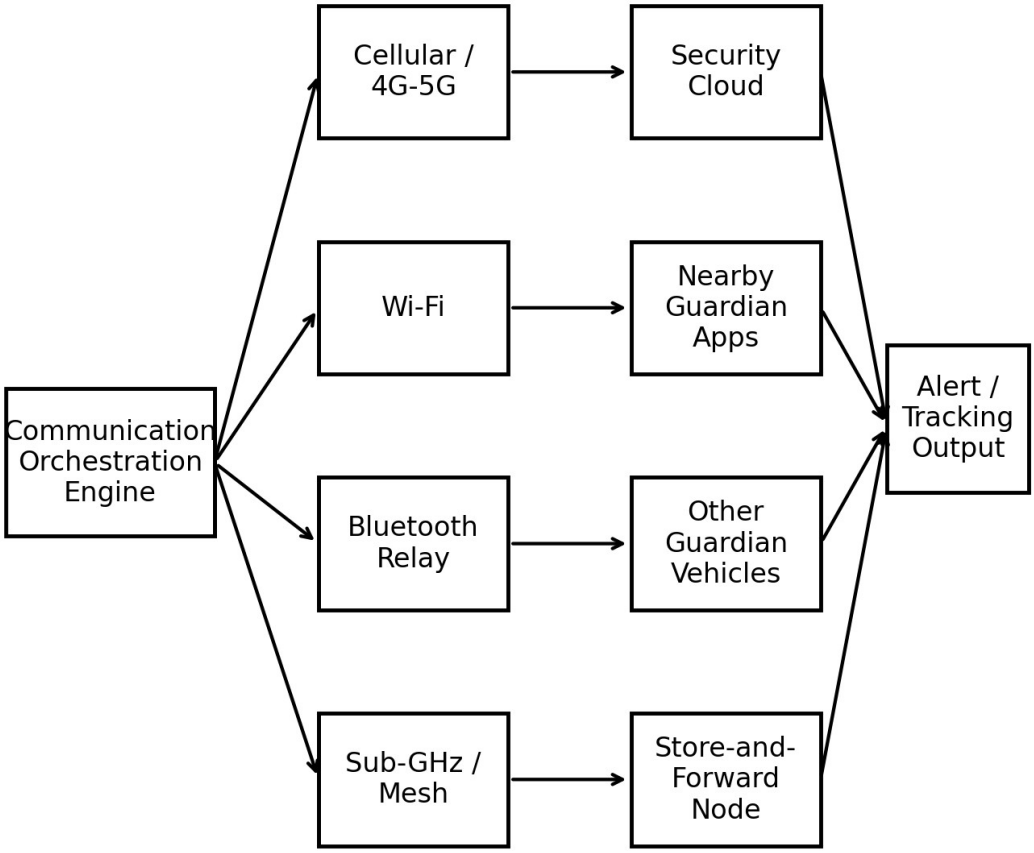


Figure 3 - Artificial intelligence detection pipeline.

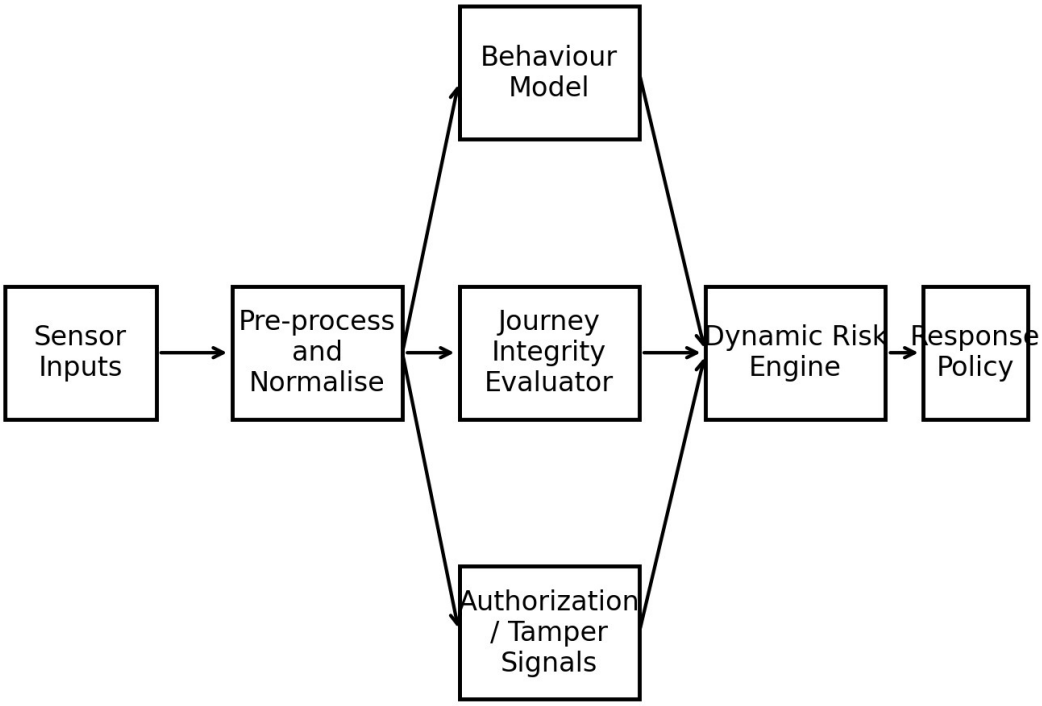


Figure 11 - Startup authorization, protected lockout, and authenticated reactivation flow.

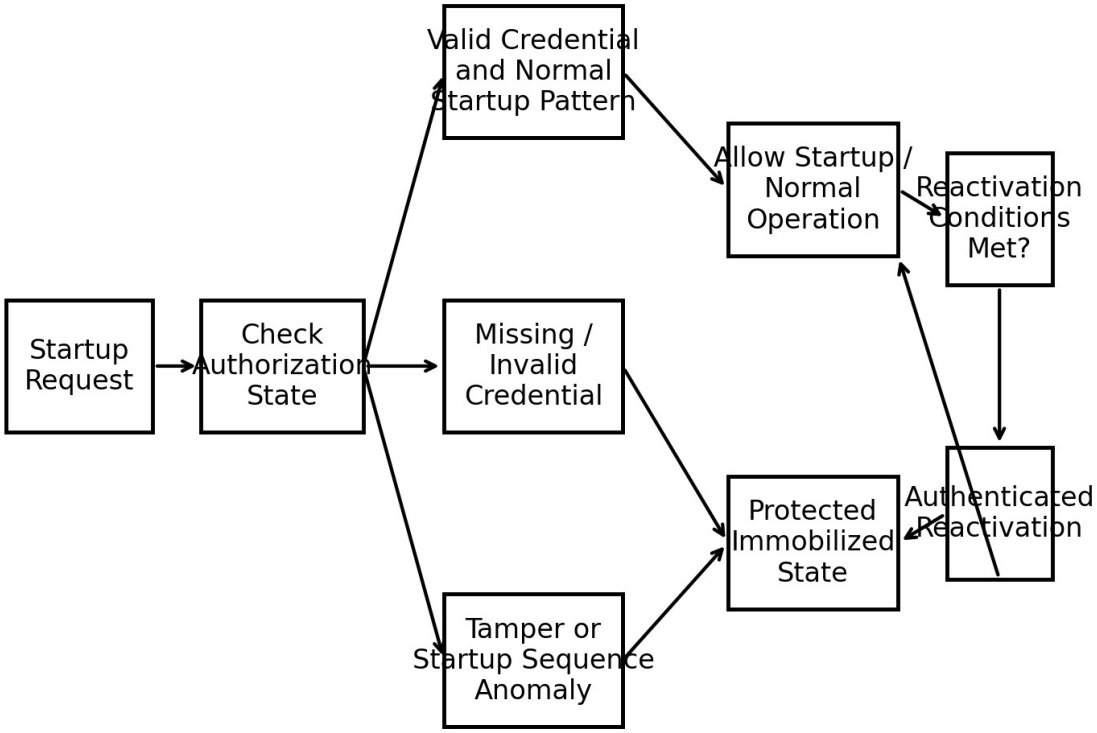


Figure 12 - Predictive threat intelligence pipeline.

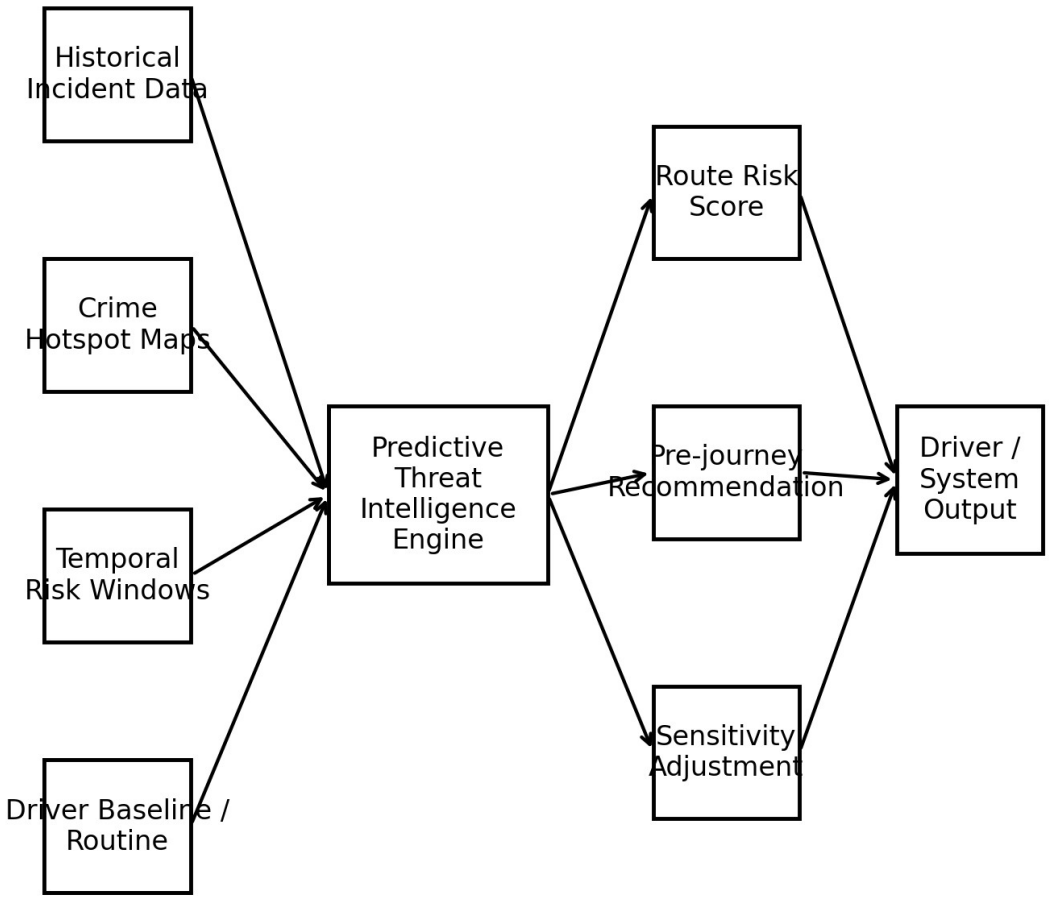


Figure 13 - Tamper-resilient evidence preservation and forensic reconstruction workflow.

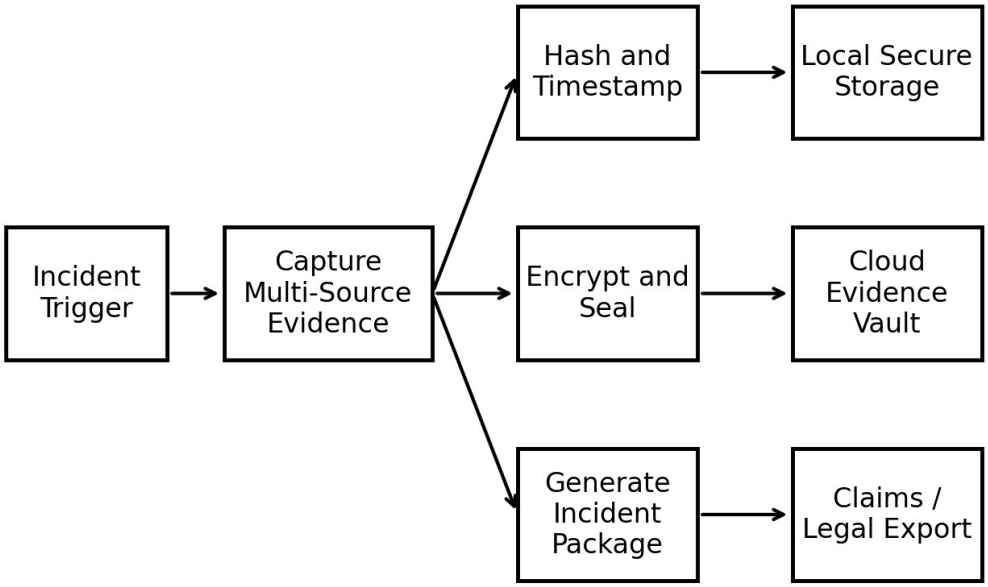
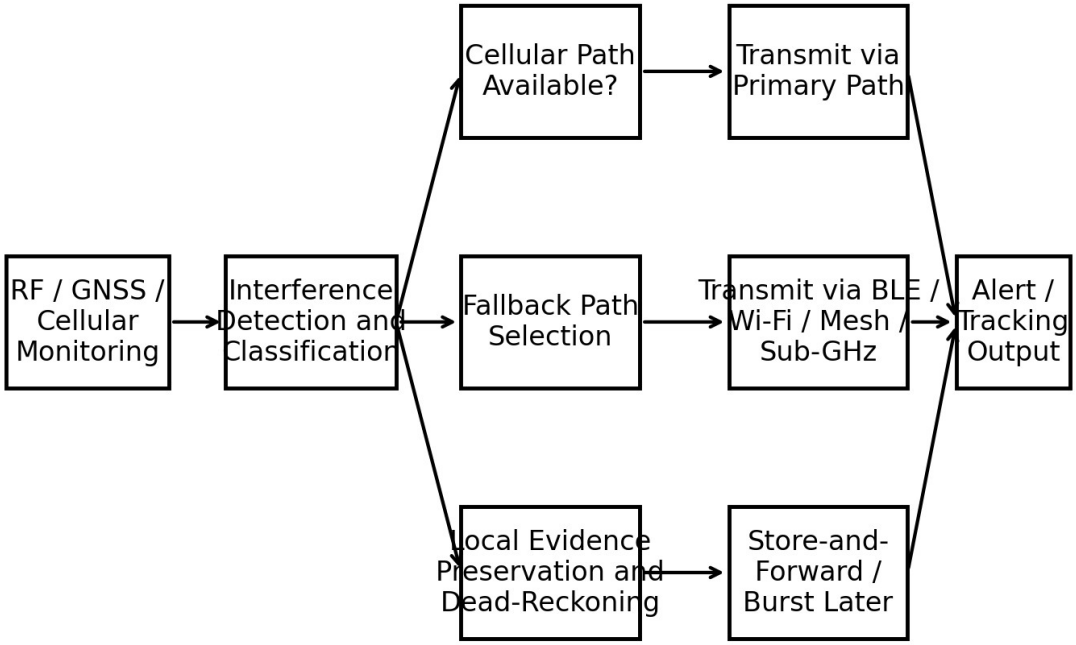


Figure 14 - Anti-jamming communication resilience workflow.



End of Specification